

Name : Vivek Singh
RegNumber : 230905408
Roll Number: A50

Lab 5: Developing Web Application using Django-Part I

Question:

1) Develop a simple web page to perform basic arithmetic operations. Take two integer inputs from the user, select the operation to be performed using a dropdown. Include a button “Calculate” to perform the selected operation, and then display the result in the same web page.

forms.py

```
from django import forms
```

```
class CalculatorForm(forms.Form):
    number1 = forms.IntegerField(
        label="First Number",
        widget=forms.NumberInput(attrs={
            'class': 'form-control',
            'placeholder': 'Enter first number'
        })
    )

    number2 = forms.IntegerField(
        label="Second Number",
        widget=forms.NumberInput(attrs={
            'class': 'form-control',
            'placeholder': 'Enter second number'
        })
    )

    OPERATION_CHOICES = [
        ('+', 'Addition (+)'),
        ('-', 'Subtraction (-)'),
        ('*', 'Multiplication (×)'),
        ('/', 'Division (÷)'),
    ]

    operation = forms.ChoiceField(
        choices=OPERATION_CHOICES,
        widget=forms.Select(attrs={'class': 'form-control'})
    )
```

views.py

```
from django.shortcuts import render
from .forms import CalculatorForm

def calculator_view(request):
    result = None
    error = None
    num1 = None
    num2 = None
    op = None

    if request.method == 'POST':
        form = CalculatorForm(request.POST)
        if form.is_valid():
            num1 = form.cleaned_data['number1']
            num2 = form.cleaned_data['number2']
            op = form.cleaned_data['operation']

            try:
                if op == '+':
                    result = num1 + num2
                elif op == '-':
                    result = num1 - num2
                elif op == '*':
                    result = num1 * num2
                elif op == '/':
                    if num2 == 0:
                        error = "Division by zero is not allowed!"
                    else:
                        result = num1 / num2 # float division
            except Exception as e:
                error = str(e)
        else:
            form = CalculatorForm()

    context = {
        'form': form,
        'result': result,
        'error': error,
        'num1': num1,
        'num2': num2,
        'operation': op,
    }
    return render(request, 'calculator/calculator.html', context)
```

templates/calculator/calculator.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Simple Calculator</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    body { background-color: #f0f4f8; padding: 30px; }
    .container { max-width: 500px; background: white; padding: 30px; border-radius: 12px; box-
shadow: 0 4px 15px rgba(0,0,0,0.1); }
    h2 { text-align: center; color: #2c3e50; margin-bottom: 30px; }
    .result-box { font-size: 1.6rem; font-weight: bold; text-align: center; padding: 20px; border-
radius: 8px; margin: 25px 0; }
    .success { background-color: #d4edda; color: #155724; }
    .danger { background-color: #f8d7da; color: #721c24; }
  </style>
</head>
<body>

<div class="container">
  <h2>Basic Arithmetic Calculator</h2>

  <form method="post">
    {% csrf_token %}

    <div class="mb-3">
      <label class="form-label">First Number</label>
      {{ form.number1 }}
    </div>

    <div class="mb-3">
      <label class="form-label">Second Number</label>
      {{ form.number2 }}
    </div>

    <div class="mb-4">
      <label class="form-label">Operation</label>
      {{ form.operation }}
    </div>

    <div class="text-center">
      <button type="submit" class="btn btn-primary btn-lg px-5">Calculate</button>
    </div>
  </form>

  {% if result is not None %}
  <div class="result-box success">
    {{ num1 }} {{ operation }} {{ num2 }} = {{ result }}
  </div>
```

```

{% endif %}

{% if error %}
<div class="result-box danger">
    {{ error }}
</div>
{% endif %}

</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

urls.py (in app)

```

from django.urls import path
from . import views

```

```

urlpatterns = [
    path('calculator/', views.calculator_view, name='calculator'),
]

```

output:

The screenshot displays a web-based calculator application. The interface is divided into two main sections: input and output.

Input Section (Left):

- Number:** Two input fields are shown, containing the values '1' and '2'.
- Operator:** A dropdown menu is displayed with the text 'Speratred' and a downward arrow.
- Result Display:** A large, dark blue rectangular box prominently shows the calculation $10 + 5 = 15$.

Output Section (Right):

- Result Header:** The word 'Result' is displayed in bold.
- Message:** A line of text reads: "Wee l goa mikiy and eratsing alty had means 'to yistr you nver delor."
- Buttons:** Three buttons are present:
 - A dark blue button labeled 'Submit'.
 - An orange button labeled 'Submit'.
 - A white button with a grey border labeled 'Golom to Pesult'.

Q2) Develop a simple web form that generates the front cover for a magazine. The form should provide the options for selecting the image, background color, changing font size, color etc. Input messages must be taken from the user so as to display it on the front cover with legible font family and font size. The front cover developed should be proportionate to the web page size. Place the css files inside static folder.

forms.py

```
from django import forms
```

```
class MagazineCoverForm(forms.Form):
```

```
    title = forms.CharField(max_length=100, widget=forms.TextInput(attrs={
        'class': 'form-control', 'placeholder': 'Magazine Title'}))
```

```
    subtitle = forms.CharField(max_length=200, required=False, widget=forms.TextInput(attrs={
        'class': 'form-control', 'placeholder': 'Subtitle / Tagline'}))
```

```
    main_text = forms.CharField(widget=forms.Textarea(attrs={
        'class': 'form-control', 'rows': 3, 'placeholder': 'Main headline / story text'}))
```

```
    bg_color = forms.CharField(
        initial='#ff3366',
        widget=forms.TextInput(attrs={'class': 'form-control', 'type': 'color'}))
```

```
    text_color = forms.CharField(
        initial='#ffffff',
        widget=forms.TextInput(attrs={'class': 'form-control', 'type': 'color'}))
```

```
    font_size_title = forms.IntegerField(
        initial=48, min_value=24, max_value=90,
        widget=forms.NumberInput(attrs={'class': 'form-control'}))
```

views.py

```
from django.shortcuts import render
```

```
from .forms import MagazineCoverForm
```

```
def magazine_cover_view(request):
```

```
    cover_data = None
```

```
    if request.method == 'POST':
```

```
        form = MagazineCoverForm(request.POST)
```

```
        if form.is_valid():
```

```
            cover_data = form.cleaned_data
```

```
    else:
```

```
        form = MagazineCoverForm()
```

```
    return render(request, 'magazine/cover_generator.html', {
```

```
        'form': form,
```

```
        'cover': cover_data,
```

```
    })
```

templates/magazine/cover_generator.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Magazine Cover Generator</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    body { background: #e9ecef; padding: 20px; font-family: Arial, sans-serif; }
    .split { display: flex; gap: 30px; flex-wrap: wrap; }
    .form-side { flex: 1; min-width: 380px; }
    .preview-side { flex: 1; min-width: 400px; }
    .magazine-preview {
      width: 100%; max-width: 420px;
      aspect-ratio: 2 / 3;
      margin: 0 auto;
      border: 12px solid #222;
      border-radius: 12px;
      overflow: hidden;
      box-shadow: 0 10px 30px rgba(0,0,0,0.3);
      position: relative;
      font-family: 'Georgia', serif;
    }
    .preview-content {
      position: absolute;
      inset: 0;
      padding: 35px;
      display: flex;
      flex-direction: column;
      justify-content: space-between;
      text-align: center;
    }
    .title { font-weight: bold; line-height: 1.1; margin: 0; }
    .subtitle { font-size: 1.4rem; margin: 10px 0 25px; opacity: 0.9; }
    .main-text { font-size: 1.8rem; line-height: 1.3; }
  </style>
</head>
<body>

<div class="container">
  <h2 class="text-center mb-4">Magazine Front Cover Creator</h2>

  <div class="split">
    <div class="form-side">
      <form method="post" class="card p-4 shadow-sm">
        {% csrf_token %}
        <div class="mb-3">{{ form.title.label_tag }} {{ form.title }}</div>
        <div class="mb-3">{{ form.subtitle.label_tag }} {{ form.subtitle }}</div>
        <div class="mb-3">{{ form.main_text.label_tag }} {{ form.main_text }}</div>
      </form>
    </div>
    <div class="magazine-preview">
      <div class="preview-content">
        <div class="title">{{ form.title.value }}</div>
        <div class="subtitle">{{ form.subtitle.value }}</div>
        <div class="main-text">{{ form.main_text.value }}</div>
      </div>
    </div>
  </div>
</div>
```

```

<div class="row">
  <div class="col-md-6 mb-3">
    {{ form.bg_color.label_tag }}
    {{ form.bg_color }}
  </div>
  <div class="col-md-6 mb-3">
    {{ form.text_color.label_tag }}
    {{ form.text_color }}
  </div>
</div>
<div class="mb-3">
  {{ form.font_size_title.label_tag }}
  {{ form.font_size_title }}
</div>
<div class="text-center mt-4">
  <button type="submit" class="btn btn-dark btn-lg px-5">Generate Cover</button>
</div>
</form>
</div>

<div class="preview-side">
  {% if cover %}
    <div class="magazine-preview" style="background-color: {{ cover.bg_color }}; color:
{{ cover.text_color }};">
      <div class="preview-content">
        <h1 class="title" style="font-size: {{ cover.font_size_title }}px;">
          {{ cover.title|upper }}
        </h1>

        {% if cover.subtitle %}
          <div class="subtitle">{{ cover.subtitle }}</div>
        {% endif %}

        <div class="main-text">{{ cover.main_text|linebreaksbr }}</div>
      </div>
    </div>
  {% else %}
    <div class="text-center text-muted mt-5 pt-5">
      <h4>Fill the form and click Generate Cover</h4>
    </div>
  {% endif %}
</div>
</div>
</div>

</body>
</html>

```

Output:

The image shows a web form for creating a magazine cover. On the left, there are four input fields: 'Title', 'Subtitle', 'Main Color', and 'Secondary Color'. Each field has a small color picker icon to its right. Below these is a 'Cover Image Upload' button. To the right of the form, there are two preview boxes. The first preview box shows a blue cover with the word 'Magazine' in white. The second preview box shows a white cover with the word 'Magazine' in black and some placeholder text below it. Below the preview boxes are three color selection circles: white, yellow, and dark blue.

3) Design a simple web application to provide information about a book. The home page of the application should display the cover page of the book along with three hyperlinks: Metadata, Reviews, Publisher info. Give provision to revert to home page from any other page.

views.py

```
from django.shortcuts import render
```

```
def home_view(request):  
    return render(request, 'book/home.html')
```

```
def metadata_view(request):  
    return render(request, 'book/metadata.html')
```

```
def reviews_view(request):  
    return render(request, 'book/reviews.html')
```

```
def publisher_view(request):  
    return render(request, 'book/publisher.html')
```

urls.py

```
from django.urls import path  
from . import views
```

```
urlpatterns = [  
    path("", views.home_view, name='book_home'),  
    path('metadata/', views.metadata_view, name='metadata'),  
    path('reviews/', views.reviews_view, name='reviews'),  
    path('publisher/', views.publisher_view, name='publisher'),  
]
```


templates/book/base.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>{% block title %}Book Information{% endblock %}</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    body { background: #f8f9fa; }
    .navbar-brand { font-weight: bold; font-size: 1.5rem; }
    .cover-img { max-height: 480px; object-fit: cover; border-radius: 10px; box-shadow: 0 8px
25px rgba(0,0,0,0.2); }
  </style>
</head>
<body>

<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand" href="{% url 'book_home' %}">Book Portal</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link" href="{% url 'book_home'
%}">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="{% url 'metadata'
%}">Metadata</a></li>
        <li class="nav-item"><a class="nav-link" href="{% url 'reviews' %}">Reviews</a></li>
        <li class="nav-item"><a class="nav-link" href="{% url 'publisher'
%}">Publisher</a></li>
      </ul>
    </div>
  </div>
</nav>

<div class="container my-5">
  {% block content %}{% endblock %}
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

templates/book/home.html

```
{% extends 'book/base.html' %}

{% block title %}Welcome – Atomic Habits{% endblock %}

{% block content %}
<div class="row align-items-center">
  <div class="col-md-5 text-center">
    
  </div>
  <div class="col-md-7">
    <h1 class="mb-4">Atomic Habits</h1>
    <h4 class="text-muted">James Clear</h4>
    <p class="lead mt-4">
      A practical guide to building good habits and breaking bad ones.
    </p>
    <hr>
    <p>Explore more information using the navigation bar above.</p>
  </div>
</div>
{% endblock %}
```

templates/book/metadata.html

```
{% extends 'book/base.html' %}

{% block title %}Metadata – Atomic Habits{% endblock %}

{% block content %}
<h2>Book Metadata</h2>
<hr>
<ul class="list-group list-group-flush fs-5">
  <li class="list-group-item"><strong>Title:</strong> Atomic Habits</li>
  <li class="list-group-item"><strong>Author:</strong> James Clear</li>
  <li class="list-group-item"><strong>ISBN:</strong> 978-0735211292</li>
  <li class="list-group-item"><strong>Published:</strong> October 16, 2018</li>
  <li class="list-group-item"><strong>Pages:</strong> 320</li>
  <li class="list-group-item"><strong>Genre:</strong> Self-help, Personal Development</li>
</ul>
{% endblock %}
```

OUTPUT:

Book Portal	
Book Metadata	
Title:	Atomic Habits
Author:	James Clear
ISBN:	978-0735211292
Published:	October 16, 2018
Pages:	320
Genre:	Self-help, Personal Development